

```
In [1]: # Library Import
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

```
In [4]: # Dataset Load
data = pd.read_csv("C:/Users/hanmant/Desktop/sdAssignment/Assignment7/mnist_train.csv")
data.head()
```

```
Out[4]:
```

	label	1x1	1x2	1x3	1x4	1x5	1x6	1x7	1x8	1x9	...	28x19	28x20	28x21	28x22
0	5	0	0	0	0	0	0	0	0	0	...	0	0	0	0
1	0	0	0	0	0	0	0	0	0	0	...	0	0	0	0
2	4	0	0	0	0	0	0	0	0	0	...	0	0	0	0
3	1	0	0	0	0	0	0	0	0	0	...	0	0	0	0
4	9	0	0	0	0	0	0	0	0	0	...	0	0	0	0

5 rows × 785 columns



```
In [5]: # Dataset Information
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 60000 entries, 0 to 59999
Columns: 785 entries, label to 28x28
dtypes: int64(785)
memory usage: 359.3 MB
```

```
In [6]: # Statistical Summary
data.describe()
```

```
Out[6]:
```

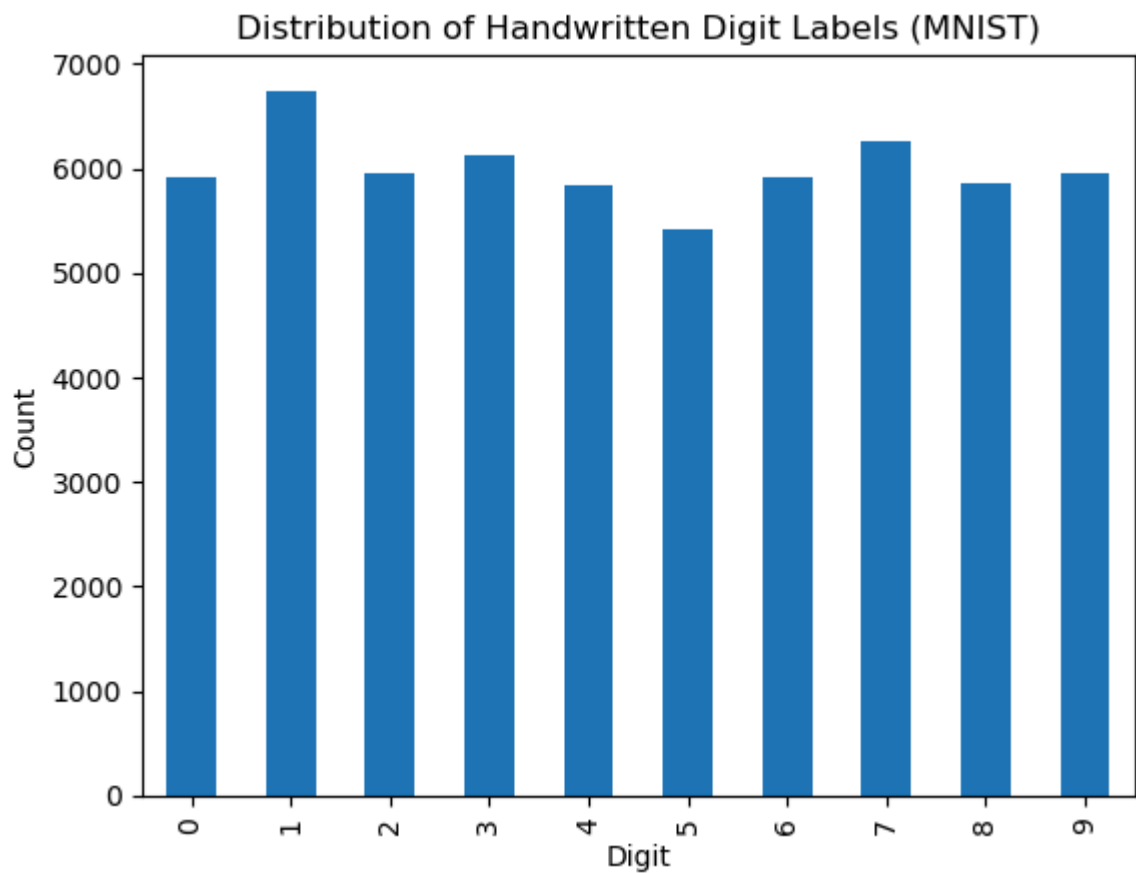
	label	1x1	1x2	1x3	1x4	1x5	1x6	1x7	1x8
count	60000.000000	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0	60000.0
mean	4.453933	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
std	2.889270	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
min	0.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
25%	2.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
50%	4.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
75%	7.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
max	9.000000	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0

8 rows × 785 columns



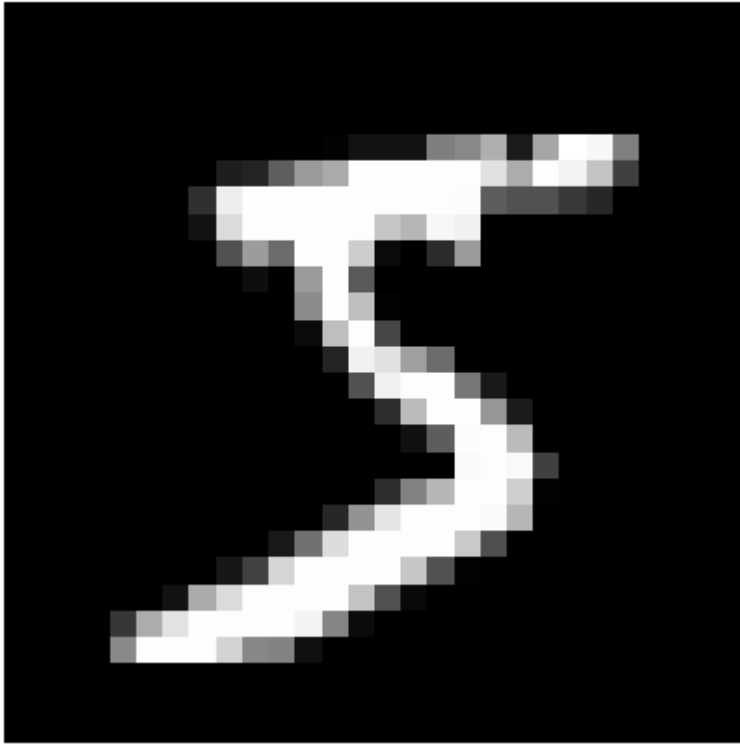
```
In [7]: # Distribution of Digit Labels (Visualization)
data['label'].value_counts().sort_index().plot(kind='bar')
```

```
plt.title("Distribution of Handwritten Digit Labels (MNIST)")
plt.xlabel("Digit")
plt.ylabel("Count")
plt.show()
```



```
In [8]: # digit image visualize
image = data.iloc[0,1:].values
image = image.reshape(28,28)
plt.imshow(image, cmap='gray')
plt.title(f"Label : {data.iloc[0,0]}")
plt.axis("off")
plt.show()
```

Label : 5



In []: